

Ionization rates near the end of reionization

Global budget per Hubble time and photoionization rate at $z = 5.5$

At $z = 5.5$, the observational benchmarks considered here are approximately **5 ionizing photons per hydrogen nucleus per Hubble time** for the global photon supply, and $\Gamma_{\text{HI}} \simeq 3.4 \times 10^{-13} \text{ s}^{-1}$ per neutral hydrogen atom. The conventional recombination-maintenance requirement is lower. These values retain the cited papers’ assumptions; they do not select a reionization history for the research project.

1. Global budget per hydrogen nucleus per Hubble time

Define the Hubble time at the epoch of interest and the dimensionless photon-supply rate as

$$t_H(z) \equiv \frac{1}{H(z)}, \quad \mathcal{N}_H(z) \equiv \frac{\dot{n}_{\text{ion}}(z)}{\bar{n}_{\text{H}}} t_H(z). \quad (1)$$

Here $\dot{n}_{\text{ion}}$ is the ionizing-photon emissivity escaping into the intergalactic medium (IGM), and $\bar{n}_{\text{H}}$ counts *all hydrogen nuclei*; both densities use consistent comoving units.

Observationally inferred supply. Gaikwad et al. [1], Table 3, report

$$\dot{n}_{\text{ion}}(5.5) = (0.587_{-0.149}^{+0.417}) \times 10^{51} \text{ photons s}^{-1} \text{ cMpc}^{-3}. \quad (2)$$

Using their cosmology ($h = 0.678$, $\Omega_m = 0.308$, $\Omega_\Lambda = 0.692$, $\Omega_b = 0.0482$, and helium mass fraction $Y = 0.24$) gives $t_H(5.5) = 1.56 \text{ Gyr}$ and

$$\boxed{\mathcal{N}_H(5.5) = 5.2_{-1.3}^{+3.7} \text{ photons per H nucleus per Hubble time.}} \quad (3)$$

The 1σ uncertainties are propagated from the published emissivity. The inference uses an absorption-limited radiation field, source spectral index $\alpha_s = 2.0 \pm 0.6$, and absorber-distribution slope $\beta_{\text{HI}} = 1.3 \pm 0.05$ [1, §6.4]. Converting the escaping photons into actual hydrogen ionizations requires accounting for their absorption and any secondary ionizations.

Conventional maintenance requirement. A commonly used prescription assumes $C_{\text{HII}} = 3$, $T = 20,000 \text{ K}$, case-B recombination, and an electron contribution from singly ionized helium [2]. The numerical approximation [3, Eq. (3)]

$$t_{\text{rec}} \simeq 0.93 \text{ Gyr} \left(\frac{C_{\text{HII}}}{3} \right)^{-1} \left(\frac{T}{2 \times 10^4 \text{ K}} \right)^{0.7} \left(\frac{1+z}{7} \right)^{-3} \quad (4)$$

gives $t_{\text{rec}}(5.5) \simeq 1.16 \text{ Gyr}$. Using Robertson et al.’s cosmology to evaluate t_H therefore yields

$$\boxed{\mathcal{N}_{H,\text{crit}}(5.5) = \frac{t_H}{t_{\text{rec}}} \simeq 1.3 \text{ ionizations per H nucleus per Hubble time.}} \quad (5)$$

This replaces recombinations in an already ionized IGM. Higher inferred clumping factors increase the requirement; Davies et al. [4] infer $C \approx 12$, normalized using case-B recombination at 10^4 K .

Meaning of the time unit. “Per Hubble time” is an instantaneous rate normalization, not a photon count accumulated over a Hubble time of evolving emissivity. Here t_H means $1/H(z)$, not the present-day Hubble time or the age of the Universe.

2. Photoionization rate per neutral hydrogen atom

At the same redshift, Gaikwad et al. [1, Table 3] report

$$\Gamma_{\text{HI}}(5.5) = \left(3.44^{+2.19}_{-1.30}\right) \times 10^{-13} \text{ s}^{-1}. \quad (6)$$

The central value corresponds to a photoionization timescale

$$\Gamma_{\text{HI}}^{-1} \simeq 92,000 \text{ yr} \quad (7)$$

for a neutral atom exposed to that background. The rate is inferred from Ly α -forest observations through the authors' simulation modeling; the quoted uncertainties are 1σ .

The normalization differs from the global budget because Γ_{HI} acts on **neutral atoms**. Locally, the photoionization-event rate per total hydrogen nucleus is $x_{\text{HI}}\Gamma_{\text{HI}}$, where x_{HI} is the neutral fraction. A global value requires the appropriate spatial average; one cannot generally replace this average by the product of separately averaged quantities.

What these benchmarks establish

Quantity at $z = 5.5$	Benchmark
Inferred escaping-photon supply per H nucleus per t_H	$5.2^{+3.7}_{-1.3}$
Conventional maintenance ionizations per H nucleus per t_H	$\simeq 1.3$
Photoionization rate per neutral H atom	$(3.44^{+2.19}_{-1.30}) \times 10^{-13} \text{ s}^{-1}$

Completion by $z \simeq 5.5$ still requires an ionization history. In the conventional photon-budget model [2, 4],

$$\frac{dQ_{\text{HII}}}{dt} = \frac{\dot{n}_{\text{ion}}}{\bar{n}_{\text{H}}} - \frac{Q_{\text{HII}}}{t_{\text{rec}}}. \quad (8)$$

The maintenance benchmark follows by setting $Q_{\text{HII}} = 1$ and $dQ_{\text{HII}}/dt = 0$. Reaching that state also depends on the earlier photon supply, initial ionized fraction, and recombination history. Neither the maintenance benchmark nor the measured Γ_{HI} alone establishes that enough earlier ionizations occurred.

References

- [1] P. Gaikwad et al. (2023). *Measuring the photo-ionization rate, neutral fraction and mean free path of HI ionizing photons at $4.9 \leq z \leq 6.0$ from a large sample of XShooter and ESI spectra*. Table 3 and §6.4. [arXiv:2304.02038](#).
- [2] B. E. Robertson et al. (2015). *Cosmic Reionization and Early Star-Forming Galaxies: A Joint Analysis of New Constraints from Planck and the Hubble Space Telescope*. Eq. (4) and adopted recombination prescription. [arXiv:1502.02024](#).
- [3] M. R. Lovell et al. (2018). *ETHOS—an effective theory of structure formation: predictions for the high-redshift Universe—abundance of galaxies and reionization*. Eq. (3). [arXiv:1711.10497](#).
- [4] F. B. Davies, S. E. I. Bosman, and S. R. Furlanetto (2024). *The Predicament of Absorption-dominated Reionization II: Observational Estimate of the Clumping Factor at the End of Reionization*. [arXiv:2406.18186](#).